LiquidityPool: Game-Theoretic Analysis of Stakeholder Revenue in Ranking-Dependent DeFi

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Abstract

Ranking-dependent DeFi protocols create participation barriers for small-volume stakeholders due to competitive disadvantages based on fund size rankings. Our analysis reveals that brokers with higher fund balances achieve better revenue performance, disadvantaging smaller investors. We introduce LiquidityPool, a fund aggregation protocol that enables collective participation in ranking-dependent DeFi environments by pooling resources from multiple stakeholders. Through game-theoretic analysis of multi-leader multi-follower Stackelberg competition, we establish theoretical results that differ from traditional competition assumptions. Our analysis reveals knife-edge equilibrium properties where symmetric multi-party competition exists only under restrictive initial conditions. When initial asymmetry exceeds a critical threshold, protocols converge to a monopolistic market structure regardless of the starting configuration. We show that monopolistic concentration performs better than fragmented competition in terms of social welfare, contrary to the predictions of classical Bertrand and Cournot competition. This occurs because convex revenue functions create economies of scale that favor fund aggregation over competitive fragmentation. Experimental validation using real Ethereum blockchain data shows monopolization across different market structures. While welfare improvements are modest due to concentrated wealth distribution, LiquidityPool enables 98% of small-volume stakeholders to participate in protocols that were previously inaccessible. These findings suggest that traditional antitrust policies promoting market fragmentation may not be effective in ranking-dependent DeFi networks, where concentration enables rather than restricts efficient resource allocation and stakeholder accessibility.

CCS Concepts

• **Theory of computation** → **Algorithmic game theory and mechanism design**; • **Applied computing** → *Economics*; • **Computer systems organization** → *Peer-to-peer architectures.*

Keywords

Blockchain, Game theory, DeFi, Market mechanism, Stackelberg competition, Economic welfare

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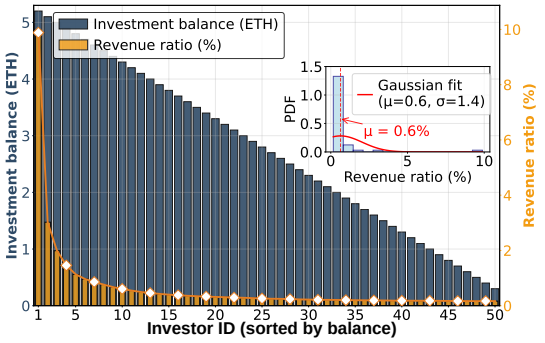


Figure 1: Correlation between investment balance and received revenue in a ranking-dependent DeFi protocol. Experimental results using a representative implementation show 50 investors, sorted by investment balance in descending order. Their individual investment balances (left-hand y-axis) decrease systematically from approximately 5 ETH to near 0 ETH. The corresponding revenue ratios (right-hand y-axis) demonstrate that investors with higher initial investment balances achieve disproportionately higher revenue ratios. We also show the probability density function (PDF) of revenue ratios with Gaussian fit parameters ($\mu = 0.6\%$, $\sigma = 1.4\%$). This validates the ranking-dependent competitive structure, in which investment balance directly correlates with earning potential, creating systematic barriers for small-volume investors.

1 Introduction

Decentralized Finance (DeFi) protocols often create systemic advantages for larger investors through mechanisms such as gas-cost amortization, capital-efficiency thresholds, and preferential access to allocations. These advantages compound over time, raising questions about accessibility and market concentration in systems designed for decentralization. Understanding the competitive dynamics of such position-sensitive protocols requires new theoretical frameworks beyond traditional market competition models.

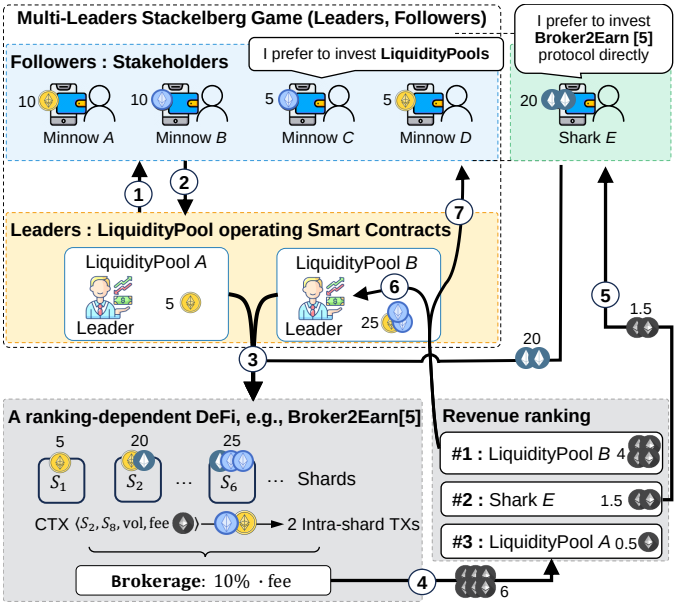


Figure 2: Overview of LiquidityPool, which enables minnows (investors/stakeholders with small-volume funds) to participate in ranking-dependent DeFi. LiquidityPool operates as a Stackelberg game with pool operators as leaders and stakeholders as followers. Step ①: LiquidityPool operators announce their management strategies to attract investors. Step ②: Minnows observe options and make investment decisions. Step ③: The staked tokens from LiquidityPools and Shark *E* are fed into the ranking-dependent DeFi protocol. Step ④: The staked tokens offer liquidity for the underlying DeFi. Brokerages are returned to all investors. Step ⑤: Individual investors (e.g., Shard *E*) receive revenue directly from the underlying DeFi. Step ⑥: LiquidityPools get revenue according to their fund sizes and management strategies. Step ⑦: All minnows receive revenue from their investment choices in LiquidityPools.

To address blockchain scalability constraints limiting DeFi expansion, sharding protocols enable parallel transaction processing through network partitioning [11, 18]. BrokerChain [8, 10] introduced specialized broker accounts to facilitate cross-shard transactions while maintaining atomicity, subsequently inspiring DeFi protocols that leverage broker mechanisms for liquidity staking. Ranking-dependent protocols, exemplified by Broker2Earn [5], exhibit competitive dynamics where investors with higher fund balances consistently achieve disproportionately higher revenue rates. Fig. 1 demonstrates this correlation in experimental implementation with 50 investors: revenue ratios exhibit a strong positive correlation with fund rankings, validating the ranking-dependent competitive structure where market position determines earning potential. Formally, we characterize such systems as *ranking-dependent* when investor revenue $f(D, r)$ exhibits (1) convex returns to scale ($\partial^2 f/\partial D^2 > 0$) and (2) position effects where larger fund holders capture better effective market positions r, yielding superlinear advantages. This emergent property, arising from transaction-allocation dynamics, creates systematic participation barriers for smaller stakeholders even when protocols do not explicitly encode ranking mechanisms. When fund distributions follow power-law patterns typical of DeFi markets, investors in the bottom 90th percentile by stake size earn negligible returns. LiquidityPool demonstrates that fund aggregation improves accessibility: experiments show participation rates increasing from 6% (direct involvement) to 100% (pooled participation) across all stakeholder segments. The competitive dynamics underlying these outcomes differ from traditional market structures.

We propose *LiquidityPool*, a fund-aggregation protocol that enables collective participation in ranking-dependent environments, where multiple stakeholders pool resources to act as unified entities, overcoming individual funding limitations through collaborative investment strategies. Fig. 2 illustrates the system architecture where small-volume investors (a.k.a. minnows) aggregate funds through LiquidityPools to achieve competitive rankings in the underlying protocol, enabling collective participation that would be infeasible individually. Our game-theoretic analysis using Stackelberg competition [17] reveals dynamics that differ from classical oligopoly predictions. Unlike Bertrand [2] or Cournot [6] competition, where symmetric equilibria emerge robustly, ranking-dependent systems exhibit knife-edge equilibria existing only under restrictive initial symmetry conditions, with inevitable monopolization when asymmetries exceed critical thresholds. We show that monopolistic concentration achieves higher social welfare than fragmented competition, because convex revenue functions favor aggregation, in contrast to traditional antitrust frameworks that presume market fragmentation improves efficiency.

Our contributions are summarized as follows.

- **Fund aggregation protocol design**: We introduce LiquidityPool with formal game-theoretic analysis demonstrating Nash equilibria existence [13, 14] and characterizing strategic behaviors of fund managers and stakeholders in ranking-dependent environments.
- **Knife-edge equilibrium analysis**: We discover that ranking-dependent competition exhibits fundamental instability where symmetric equilibria exist only under restrictive initial conditions, contrasting with classical competition models and revealing inevitable monopolization dynamics.
- **Welfare analysis**: We prove that monopolistic concentration outperforms fragmented competition in social welfare, differing from traditional antitrust frameworks where market fragmentation is presumed to improve efficiency.
- **Theoretical and empirical validation**: We provide rigorous mathematical analysis using Lyapunov stability theory [12] and validate predictions through simulations using real Ethereum blockchain data from blocks 19,000,000-19,999,999.

2 Related Work

DeFi protocols have attracted research on mechanism design and market microstructure [1, 3, 4, 7, 15]. Angeris and Chitra [1] analyze constant-function market makers and establish price-oracle properties for automated market makers such as Uniswap. Capponi and Jia [4] examine adoption dynamics in blockchain-based decentralized exchanges, while Gudgeon et al. [7] identify systemic

risks in DeFi lending protocols. Qin et al. [15] quantify miner extractable value in DeFi ecosystems. These works focus on settings where competition among protocols improves efficiency, typically assuming identical strategies yield identical outcomes for investors.

Classical competition models provide the foundation for understanding market outcomes in different strategic settings [2, 6, 17]. While Bertrand competition predicts marginal-cost pricing converging to zero profits under perfect competition, Cournot competition allows positive profits through quantity restrictions where multiple firms coexist [16]. Both models assume firms with identical technologies achieve identical unit revenues. Justitia [20] proposes a subsidy-based incentive mechanism that improves fairness (and reduces latency) for cross-shard transactions in sharded blockchains. Our ranking-dependent setting differs fundamentally because market positions create endogenous differentiation among otherwise identical investors, with advantages compounding rather than being competed away.

BrokerChain [8, 10] introduced broker-mediated cross-shard transactions in sharded blockchain systems, implemented on the BlockEmulator simulation platform [9], creating transaction allocation mechanisms based on broker performance. Building on this infrastructure, protocols like BrokerFi [19] and Broker2Earn [5] leverage broker mechanisms for DeFi applications, including liquidity staking. Our analysis reveals that these systems exhibit competitive dynamics in which fund aggregation and market position jointly determine returns, unlike traditional settings where competition dissipates initial advantages. This requires examining how ranking-dependent revenue structures affect the existence of equilibria and the stability of welfare outcomes.

3 System Model and Game Formulation

3.1 System Model

We model strategic interactions in ranking-dependent DeFi environments using discrete time intervals, called epochs, that represent decision-making rounds in which LiquidityPools set management fees, stakeholders choose participation strategies, and revenue is allocated based on cross-shard transaction processing.

Market investors. Our protocol involves LiquidityPools $\mathcal{B} = \{1, 2, \ldots, b, \ldots\}$ providing fund aggregation services and stakeholders $\mathcal{V} = \{1, 2, \ldots, v, \ldots\}$ seeking revenue optimization. Stakeholders partition into $\mathcal{M}$ (direct investors of Broker2Earn) and $\mathcal{H}$ (investor of LiquidityPool) where $\mathcal{V} = \mathcal{M} \cup \mathcal{H}$.

Decision Variables. For each stakeholder $v \in \mathcal{V}$ and LiquidityPool $b \in \mathcal{B}$, we define binary participation:

$$Y_b^v = \begin{cases} 1, & \text{if stakeholder } v \in \mathcal{V} \text{ joins LiquidityPool } b \in \mathcal{B}; \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Constraint $\sum_{b\in\mathcal{B}} Y_b^v \leq 1$ ensures each stakeholder joins at most one pool, with $\sum_{b\in\mathcal{B}} Y_b^v = 0$ representing direct participation. We denote $\boldsymbol{Y}_v = \{Y_b^v | b \in \mathcal{B}\}$ as stakeholder v's participation vector and $\mathbf{Y} = \{\boldsymbol{Y}_v | v \in \mathcal{V}\}$ as the complete decision space.

Each stakeholder $v \in \mathcal{V}$ possesses account balance $D_V(v) > 0$. When stakeholders join LiquidityPool b, their funds aggregate to:

$$D_B(b, \boldsymbol{\delta}) = \sum_{v\in\mathcal{V}} Y_b^v \cdot D_V(v), \quad v \in \mathcal{V}, b \in \mathcal{B}, \tag{2}$$

where $\boldsymbol{\delta} = \{\delta_1, \ldots, \delta_b, \ldots\}$ is management fee ratios. Rankings are determined by fund sizes with predetermined tie-breaking rules.

Revenue Structure. For any investor with funds D and market ranking $r \in \{1, 2, 3, \ldots\}$, revenue is:

$$f(D, r), \text{ where } \frac{\partial f}{\partial D} > 0, \frac{\partial^2 f}{\partial D^2} > 0, \text{ and } f(D, r_1) > f(D, r_2) \text{ for } r_1 < r_2. \tag{3}$$

This specification captures scale effects through convexity and ranking effects, where identical funds yield different revenues based on market position.

LiquidityPool $b \in \mathcal{B}$ with funds $D_B(b, \boldsymbol{\delta})$ and ranking r_b obtains total revenue $f(D_B(b, \boldsymbol{\delta}), r_b)$, distributed proportionally among investors. For stakeholder v joining pool b with management fee $\delta_b \in [0, 1]$, net revenue is $(1 - \delta_b) \cdot \frac{D_V(v)}{D_B(b,\boldsymbol{\delta})} \cdot f(D_B(b, \boldsymbol{\delta}), r_b)$, while direct participation yields $f(D_V(v), r_v)$ where r_v represents the stakeholder's individual ranking.

3.2 Game Formulation

We model interactions as a Stackelberg game in which LiquidityPools (leaders) set management fees before stakeholders (followers) choose their participation strategies.

Stakeholder Problem (Followers). Each stakeholder $v \in \mathcal{V}$ chooses participation to maximize net revenue $U_f^v = \sum_{b\in\mathcal{B}} Y_b^v \cdot (1 - \delta_b) \cdot \frac{D_V(v)}{D_B(b,\boldsymbol{\delta})} \cdot f(D_B(b, \boldsymbol{\delta}), r_b) + \left(1 - \sum_{b\in\mathcal{B}} Y_b^v\right) \cdot f(D_V(v), r_v)$:

$$\max_{\{Y_b^v\}} \quad U_f^v \tag{4}$$

$$\text{s.t.} \quad \sum_{b\in\mathcal{B}} Y_b^v \leq 1, \quad Y_b^v \in \{0, 1\}, \ \forall b \in \mathcal{B}, v \in \mathcal{V}. \tag{5}$$

LiquidityPool Problem (Leaders). Each LiquidityPool $b \in \mathcal{B}$ chooses management fee to maximize revenue $U_l^b = \delta_b \cdot f(D_B(b, \boldsymbol{\delta}), r_b)$, anticipating stakeholder responses:

$$\max_{\delta_b} \quad U_l^b \tag{6}$$

$$\text{s.t.} \quad \frac{\partial f}{\partial D} > 0, \frac{\partial^2 f}{\partial D^2} > 0, \quad \delta_b \in [0, 1], \ b \in \mathcal{B}, \tag{7}$$

where $D_B(b, \boldsymbol{\delta})$ and r_b depend on equilibrium stakeholder choices given fee vector $\boldsymbol{\delta}$.

Definition 1 (LiquidityPool Recruiting Game Γ). *The Stackelberg game $\Gamma = (\mathcal{B}, \mathcal{V}, \boldsymbol{\delta}, \mathbf{Y}, \{U_l^b\}_{b\in\mathcal{B}}, \{U_f^v\}_{v\in\mathcal{V}})$ where leaders $\mathcal{B}$ set management fees $\boldsymbol{\delta} = \{\delta_b | b \in \mathcal{B}\}$, followers $\mathcal{V}$ choose $\mathbf{Y} = \{\boldsymbol{Y}_v | v \in \mathcal{V}\}$ with $\boldsymbol{Y}_v = \{Y_b^v | b \in \mathcal{B}\}$, and payoffs are U_l^b for LiquidityPool $b \in \mathcal{B}$ and U_f^v for stakeholder $v \in \mathcal{V}$.*

The LiquidityPool recruiting game represents a multi-leader, multi-follower Stackelberg competition. LiquidityPool leaders independently determine management fees based on historical market data. Stakeholders then observe fee announcements and make participation decisions to maximize expected returns, creating competitive environments in which leaders anticipate follower responses when setting strategies. In contrast, followers evaluate LiquidityPool offerings under ranking-dependent revenue structures.

4 Theory Foundation

This section analyzes competition dynamics in ranking-dependent DeFi protocols, establishing stakeholder optimal responses, equilibrium existence and stability properties, and welfare implications. Our analysis reveals critical threshold phenomena that determine market concentration outcomes.

4.1 Stakeholder Response and Market Dynamics

Ranking-dependent revenue functions with convexity ($\frac{\partial^2 f}{\partial D^2} > 0$) and ranking effects create complex allocation incentives. In classical portfolio theory, diversification reduces risk, but when revenue exhibits superlinear growth in fund size and depends on market position, the optimal strategy becomes unclear. We establish that concentrated strategies dominate diversified ones, thereby justifying the use of binary participation variables throughout our analysis.

Theorem 1 (All-or-Nothing Investment Strategy). *In game Γ with ranking-dependent revenue function $f(D, r)$ satisfying Eq. (3), for any stakeholder $v \in \mathcal{V}$ and any diversified allocation strategy $\boldsymbol{\varsigma}^v = (\varsigma_0^v, \varsigma_1^v, \ldots, \varsigma_{|\mathcal{B}|}^v)$ where ς_0^v represents the fraction allocated to direct Broker2Earn participation, ς_b^v represents the fraction allocated to LiquidityPool $b \in \mathcal{B}$, with $\varsigma_0^v + \sum_{b\in\mathcal{B}} \varsigma_b^v = 1$ and $\varsigma_k^v \geq 0$, there exists a concentrated strategy $\hat{\boldsymbol{\varsigma}}^v$ such that $U_f^v(\hat{\boldsymbol{\varsigma}}^v) \geq U_f^v(\boldsymbol{\varsigma}^v)$, where the concentrated strategy allocates $D_V(v)$ entirely to the option yielding $\max\{f(D_V(v), r_v), \{h(v, \delta_b) + f(D_V(v), r_v)\}_{b\in\mathcal{B}}\}$, and $h(v, \delta_b) = (1 - \delta_b) \cdot \frac{D_V(v)}{D_B(b,\boldsymbol{\delta})} \cdot f(D_B(b, \boldsymbol{\delta}), r_b) - f(D_V(v), r_v)$.*

Theorem 1 establishes that concentration dominates diversification, with proof in Appendix A.1. Convexity creates superlinear returns: a stakeholder with 100 tokens in one pool generates more revenue than splitting 50 tokens across two pools, since $f(100, r) > f(50, r) + f(50, r')$ by strict convexity. Ranking effects amplify this advantage: concentrated positions achieve higher rankings, while diversified positions fragment across lower tiers. This justifies using binary variables $Y_b^v \in \{0, 1\}$ throughout our analysis. Without LiquidityPool aggregation, small-volume stakeholders face a fundamental tension between concentration optimality and insufficient scale for competitive rankings.

With concentration optimality, we now characterize the specific option stakeholders choose and the migration conditions.

Theorem 2 (Stakeholder Best Response). *Given LiquidityPool fee ratios $\boldsymbol{\delta}^* = \{\delta_b^* \mid b \in \mathcal{B}\}$, the optimal strategy of stakeholder $v \in \mathcal{V}$ is characterized by utility difference $h(v, \delta_b^*) = (1 - \delta_b^*) \cdot \frac{D_V(v)}{D_B(b,\boldsymbol{\delta}^*)} \cdot f(D_B(b, \boldsymbol{\delta}^*), r_b) - f(D_V(v), r_v)$, where v joins pool b (setting $Y_b^{v*} = 1$) if and only if $h(v, \delta_b^*) > 0$ and $b = \arg\max_{b'\in\mathcal{B}} h(v, \delta_{b'}^*)$, chooses direct participation (setting $Y_b^{v*} = 0$ for all b) if $h(v, \delta_b^*) \leq 0$ for all b, and is indifferent when $h(v, \delta_b^*) = 0$ at the maximizing pool.*

Theorem 2 characterizes best responses through utility differences, with proof in Appendix A.2. Stakeholders trade off management fees against aggregation benefits: joining a pool costs δ_b^* in fees but gains superior returns through scale and ranking if the pool's revenue per unit exceeds what the stakeholder achieves directly. Small fees or ranking changes trigger discrete threshold crossings, leading to discrete movements rather than smooth reallocations and creating step-function fund dynamics.

Lemma 1 (Fund Migration Properties). *For any LiquidityPool $b \in \mathcal{B}$, the aggregated fund function $D_B(b, \boldsymbol{\delta})$ exhibits: (1) $D_B(b, \boldsymbol{\delta})$ is a non-increasing step function in δ_b when other pools' fees $\boldsymbol{\delta}_{-b}$ are held constant, taking finitely many constant values with discontinuous jumps at threshold points where stakeholders' participation decisions change, and (2) cross-partial derivative $\frac{\partial^2 U_l^b}{\partial\delta_b \partial\delta_{b'}} < 0$ for all $b \neq b' \in \mathcal{B}$, indicating strategic substitutability where competitor fee increases make own fee increases more profitable.*

Lemma 1 establishes strategic substitutability differing from Bertrand competition, where competitor price increases enable undercutting ($\frac{\partial^2 \pi_b}{\partial p_b \partial p_{b'}} > 0$), with proofs in Appendix A.3. Here, competitor b' raising fees triggers migration to pool b, creating scale effects (convexity $\frac{\partial^2 f}{\partial D^2} > 0$) and ranking effects that make higher own fees δ_b more profitable. This generates positive feedback, amplifying initial advantages.

4.2 Conditional Equilibrium and Knife-Edge Dynamics

Strategic substitutability in Lemma 1 suggests initial asymmetries amplify rather than dissipate. In traditional Bertrand or Cournot competition, symmetric equilibria emerge robustly and remain stable under small perturbations. We established an equilibrium that exists but exhibits knife-edge instability, where symmetric multi-party competition requires restrictive initial symmetry conditions.

Theorem 3 (Nash Equilibrium Existence). *Given the LiquidityPool recruiting game $\Gamma = (\mathcal{B}, \mathcal{V}, \boldsymbol{\delta}, \boldsymbol{\mathcal{Y}}, \{U_l^b\}_{b\in\mathcal{B}}, \{U_f^v\}_{v\in\mathcal{V}})$, there exists at least one pure strategy Nash equilibrium $(\boldsymbol{\delta}^*, \boldsymbol{\mathcal{Y}}^*)$ such that $U_l^b(\{\delta_b^*\} \cup \boldsymbol{\delta}_{-b}^*, \boldsymbol{\mathcal{Y}}^*) \geq U_l^b(\{\delta_b\} \cup \boldsymbol{\delta}_{-b}^*, \boldsymbol{\mathcal{Y}}^*)$ for all $\delta_b \in [0, 1], b \in \mathcal{B}$, and $U_f^v(\boldsymbol{\delta}^*, \boldsymbol{\mathcal{Y}}^{v*} \cup \boldsymbol{\mathcal{Y}}_{-v}^*) \geq U_f^v(\boldsymbol{\delta}^*, \boldsymbol{\mathcal{Y}}^v \cup \boldsymbol{\mathcal{Y}}_{-v}^*)$ for all $\boldsymbol{\mathcal{Y}}^v, v \in \mathcal{V}$.*

Theorem 3 guarantees existence through Brouwer's fixed point theorem, with proof in Appendix A.4. Despite step-function discontinuities, the best-response correspondence remains upper semi-continuous on a compact strategy space. Equilibrium depends sensitively on initial allocations: symmetric conditions yield symmetric equilibria, while asymmetric allocations amplify via strategic substitutability, motivating the asymmetry parameter α.

Initial Allocation Asymmetry. When LiquidityPools deploy with identical parameters, deviations from perfect symmetry are characterized by $D_B(b, \boldsymbol{\delta}) = \frac{\sum_{v\in\mathcal{V}} D_V(v)}{|\mathcal{B}|} \times (1 + \varepsilon_b)$ where perturbations satisfy $\sum_{b\in\mathcal{B}} \varepsilon_b = 0$ and $\max_{b\in\mathcal{B}} |\varepsilon_b| \leq \frac{1-\alpha}{\alpha}$. The asymmetry parameter $\alpha \in (0, 1]$ quantifies deviation: $\alpha = 1$ represents perfect symmetry while $\alpha \to 0$ indicates severe asymmetry.

Theorem 4 (Knife-Edge Equilibrium Property). *In game Γ with asymmetry parameter $\alpha \in (0, 1]$, there exists critical threshold $\alpha^* \in (0, 1)$ such that symmetric Nash equilibria where $\delta_b^* = \delta^*$ for all $b \in \mathcal{B}$ exist when $\alpha \geq \alpha^*$ but not when $\alpha < \alpha^*$, where $\alpha^* = [1 + \frac{(|\mathcal{B}|-1)\mu-\lambda}{\lambda} \cdot \frac{\partial^2 f/\partial D^2}{\partial^2 f/\partial D\partial r}]^{-1}$, with $\lambda, \mu > 0$ as migration response parameters, $\frac{\partial^2 f}{\partial D^2}$ capturing revenue convexity, $\frac{\partial^2 f}{\partial D\partial r}$ representing ranking effects, and $|\mathcal{B}|$ denoting pool count.*

Theorem 4 establishes knife-edge properties through linearized stability analysis, with proof in Appendix A.5. The proof technique represents the core methodological contribution. We linearize fund allocation dynamics around symmetric equilibrium, defining deviation vector $\mathbf{x}(t) = (x_1(t), \ldots, x_{|\mathcal{B}|}(t))$ where $x_b(t) = D_B(b, \boldsymbol{\delta}^t) - \frac{D_{\text{total}}}{|\mathcal{B}|}$. The evolution $\frac{dx_b(t)}{dt} = \sum_{b'} A_{bb'} x_{b'}(t)$ follows a stability matrix with diagonal elements $A_{bb} = -\lambda(\frac{\partial^2 f}{\partial D^2} + \frac{\partial^2 f}{\partial D \partial r})$ (self-stabilization) and off-diagonal elements $A_{bb'} = \mu \frac{\partial^2 f}{\partial D^2}$ (competitive amplification). The largest eigenvalue $\lambda_{\max} = -\lambda \frac{\partial^2 f}{\partial D^2} + (|\mathcal{B}| - 1)\mu \frac{\partial^2 f}{\partial D^2} - \lambda \frac{\partial^2 f}{\partial D \partial r}$ determines stability: $\alpha \geq \alpha^*$ ensures $\lambda_{\max} < 0$ (decay), while $\alpha < \alpha^*$ yields $\lambda_{\max} > 0$ (divergence).

The critical threshold α^* decreases as revenue convexity ($\frac{\partial^2 f}{\partial D^2}$) strengthens economies of scale, as ranking effects ($\frac{\partial^2 f}{\partial D \partial r}$) amplify competitive advantages, and as more pools ($|\mathcal{B}|$) fragment the market. At this knife-edge equilibrium, symmetric outcomes exist only at the boundary $\alpha = \alpha^*$—infinitesimal deviations cause instability, contrasting sharply with Bertrand and Cournot models where symmetric equilibria withstand finite perturbations.

Theorem 5 (Market Structure and Monopolization). *In game Γ, Nash equilibria satisfy dichotomy: **(i)** when $\alpha < \alpha^*$, any configuration converges to monopolistic equilibrium where unique $b^* \in \mathcal{B}$ captures all funds with $D_B(b^*, \hat{\boldsymbol{\delta}}) = \sum_v D_V(v)$ and $D_B(b, \hat{\boldsymbol{\delta}}) = 0$ for $b \neq b^*$, where $b^* = \arg\max_b D_B(b, \boldsymbol{\delta}^0)$ is determined by initial allocations; **(ii)** when $\alpha \geq \alpha^*$, knife-edge symmetric equilibria exist but remain vulnerable; **(iii)** any equilibrium belongs to $\{$symmetric, monopolistic$\}$ with no stable intermediate configurations.*

Theorem 5 establishes monopolization when $\alpha < \alpha^*$, proved in Appendix A.6. Ranking-amplified feedback creates winner-takes-all dynamics in which pools that gain initial advantages improve rankings and raise fees, reinforcing concentration. Monopolist identity is path-dependent, determined by initial allocation rather than operations. Two identical pools with nearly identical initial allocations diverge: the larger becomes a monopolist while the smaller exits.

The knife-edge instability and inevitable monopolization contrast sharply with classical oligopoly predictions. In Bertrand competition, any price increase triggers undercutting that restores marginal-cost pricing. In Cournot competition, firms choose output levels that yield interior equilibria in which multiple firms coexist with positive profits, and increasing competitors' output improves welfare. Our ranking-dependent protocols exhibit opposite properties. Strategic substitutability (Lemma 1) means that competitor fee increases make the own fee increases more profitable, rather than triggering undercutting, thereby reversing Bertrand's predictions. Initial asymmetries amplify rather than dissipate, with α^* determining whether symmetry can be sustained. Increasing the number of competing pools raises α^*, shrinking the region where symmetric equilibria remain stable, in contrast to Cournot, where more competitors improve welfare. These differences stem from convex ranking-dependent revenue-generating winner-takes-all dynamics in which advantages compound rather than being competed away.

4.3 Welfare Analysis

Inevitable concentration raises welfare questions with profound policy implications. Traditional antitrust economics presumes monopolistic structures reduce welfare, but ranking-dependent revenue creates convex returns to aggregation, suggesting concentration might improve rather than harm welfare.

Theorem 6 (Welfare Analysis). *Define social welfare as $W(\boldsymbol{\delta}, \mathcal{Y}) = \sum_{b \in \mathcal{B}} f(D_B(b, \boldsymbol{\delta}), r_b) + \sum_{v \in \mathcal{M}} f(D_V(v), r_v)$ where $\mathcal{M} \subseteq \mathcal{V}$ denotes direct investors. Let W^* represent the social optimum with complete aggregation at zero fees, $\hat{W}$ denote the monopolistic equilibrium when $\alpha < \alpha^*$, and $\tilde{W}$ represent the symmetric equilibrium when $\alpha \geq \alpha^*$. Welfare outcomes satisfy $W^* > \hat{W} > \tilde{W}$, with Price of Anarchy $PoA(\alpha) = \frac{W^*}{\hat{W}} \in [1, \gamma]$ when $\alpha < \alpha^*$ and $PoA(\alpha) = \frac{W^*}{\tilde{W}}$ when $\alpha \geq \alpha^*$, where $\gamma > 1$ accounts for monopolistic exclusion.*

Theorem 6 establishes counterintuitive ordering $\hat{W} > \tilde{W}$, with proof in Appendix A.7. This differs from classical theory, where fragmentation improves welfare; here, welfare decreases as funds fragment across more pools. Convex revenue creates economies of scale where aggregating funds generates more revenue than splitting by Jensen's inequality: $f(D, r) > |\mathcal{B}| \cdot f(\frac{D}{|\mathcal{B}|}, r)$. Monopolistic concentration achieves superior rankings compared to fragmented pools with smaller allocations, further amplifying aggregation advantages. The Price of Anarchy remains modest (experimental validation shows PoA ≈ 1.008) due to concentrated wealth distribution. This suggests that traditional antitrust policies promoting fragmentation may be counterproductive in ranking-dependent protocols, where concentration enables efficient resource allocation, shifting the policy focus from promoting competition to ensuring accessibility through fee caps or progressive fee structures that preserve aggregation benefits while reducing exclusion.

5 Player Strategies and Decision Mechanisms

We present a systematic approach for modeling strategic decisions within the LiquidityPool recruiting game Γ. These algorithms represent our simulation implementation—in practice, different investors employ diverse strategies based on their objectives and market understanding. An epoch defines a discrete time period for investor decision-making and interaction.

Our simulation follows a five-step workflow: (1) prepare market data from smart contracts; (2) LiquidityPool fee determination (Algorithm 1); (3) stakeholder participation decisions (Algorithm 2); (4) execute Broker2Earn transaction allocation; (5) calculate revenue distribution for next epoch. Transaction data derives from real Ethereum blockchain networks.

5.1 LiquidityPool Decision Mechanism

The goal of LiquidityPool management is to maximize revenue through strategic fee optimization under competitive ranking-dependent conditions. To achieve this goal, Algorithm 1 executes adaptive management fee ratio (MFR) adjustments in response to market dynamics and stakeholder behavior.

Algorithm 1 is executed epoch by epoch. Let t denote each epoch of the LiquidityPool system. At the beginning of epoch t, the algorithm observes market data including LiquidityPool registry $\mathcal{B}$, Broker2Earn revenue records $\mathcal{F}_t$, and LiquidityPool fund sizes $\mathcal{D}_t$

Algorithm 1: LiquidityPool MFR Decision

Input : LiquidityPool b, Current epoch t, Previous MFR δ_b^{t-1}, Threshold k
Output : MFR δ_b^t for epoch t

1 $\mathcal{B} \leftarrow$ All LiquidityPools registered in smart contract
2 $\mathcal{F}_t \leftarrow$ B2E revenue in epoch t // *$\mathcal{F}_t(b')$ denotes the revenue of LiquidityPool $b' \in \mathcal{B}$ acting as a special broker in B2E.*
3 $\mathcal{D}_t \leftarrow \{D(b') \mid \forall b' \in \mathcal{B}\}$.
4 **if** $t = 1$ **then**
5 $\quad \mathcal{F} \leftarrow \{\mathcal{F}_t\}, \mathcal{D} \leftarrow \{\mathcal{D}_t\}$
6 $\quad$ cSucc $\leftarrow 0$, cFail $\leftarrow 0$, $\delta_b^t \leftarrow \delta_b^{t-1}$
7 $\quad \overline{\delta} \leftarrow$ the maximum limitation of MFR
8 $\quad \delta_{min} \leftarrow$ the minimum limitation of MFR
9 **else**
10 $\quad \mathcal{F} \leftarrow \mathcal{F} \cup \{\mathcal{F}_t\}, \mathcal{D} \leftarrow \mathcal{D} \cup \{\mathcal{D}_t\}$
11 $\quad \Delta\mathcal{F} \leftarrow \frac{1}{\min(t,k)} \sum_{i=\max(1,t-k+1)}^{t} (\mathcal{F}_i(b) - \mathcal{F}_{i-1}(b))$
12 $\quad \Delta\mathcal{D} \leftarrow \frac{1}{\min(t,k)} \sum_{i=\max(1,t-k+1)}^{t} (\mathcal{D}_i(b) - \mathcal{D}_{i-1}(b))$
13 $\quad$ **if** $\Delta\mathcal{F} \geq \epsilon$ *or* $\Delta\mathcal{D} \geq \epsilon$ **then**
14 $\quad\quad$ cSucc $\leftarrow$ cSucc $+ 1$, cFail $\leftarrow 0$
15 $\quad\quad$ **if** cSucc $\geq k$ **then**
16 $\quad\quad\quad \overline{\delta} \leftarrow \max(\overline{\delta}, \delta_b^{t-1})$
17 $\quad$ **else if** $\Delta\mathcal{F} \leq -\epsilon$ **then**
18 $\quad\quad$ cFail $\leftarrow$ cFail $+ 1$, cSucc $\leftarrow 0$
19 $\quad\quad$ **if** cFail $\geq k$ **then**
20 $\quad\quad\quad \overline{\delta} \leftarrow \min(\overline{\delta}, \delta_b^{t-1} \times 0.95)$
21 $\quad$ **if** $\mathcal{D}_t(b) \leq \epsilon$ **then**
22 $\quad\quad \delta_b^t \leftarrow \delta_b^{t-1} \times 0.5$
23 $\quad$ **else**
24 $\quad\quad \alpha \leftarrow 0.1 \times \log_2(\frac{\max(\{\mathcal{D}_t(b') | b' \in \mathcal{B}\})}{\mathcal{D}_t(b)+\epsilon})$
25 $\quad\quad$ upperBound $\leftarrow \overline{\delta} \times$ uniformRandom$(1 - \frac{1}{t}, 1 + \frac{1}{t})$
26 $\quad\quad \delta_b^t \leftarrow \max(\delta_{min}, \min(\delta_b^{t-1} + \alpha, \text{upperBound}))$
27 **return** δ_b^t

from smart contract queries (line 1 - 3). For the initial epoch, the algorithm initializes historical data structures $\mathcal{F}$ and $\mathcal{D}$, sets counters *cSucc* and *cFail* to zero, and establishes MFR bounds $\overline{\delta}$ and δ_{min} (lines 4-8). For subsequent epochs, the algorithm updates historical records and computes performance trends $\Delta\mathcal{F}$ and $\Delta\mathcal{D}$ over the recent window using difference-based moving averages (lines 10-11). Based on consecutive success or failure patterns tracked by counters, the dynamic upper bound $\overline{\delta}$ is adjusted: successful trends increase the bound to current MFR levels (lines 13-16), while persistent failures reduce it by 5% (lines 17-20). When the LiquidityPool fund size falls below the threshold ϵ, emergency response immediately halves the MFR to restore competitiveness (lines 21-22). Otherwise, the algorithm calculates the competitive adjustment factor α as the ratio of the current pool's fund size to the strongest competitor's fund size. It applies bounded optimization with randomized upper bound perturbation (lines 23-26). The LiquidityPool Decision Algorithm 1 continues execution throughout the protocol's operational lifetime.

Algorithm 2: Stakeholder Decision Algorithm

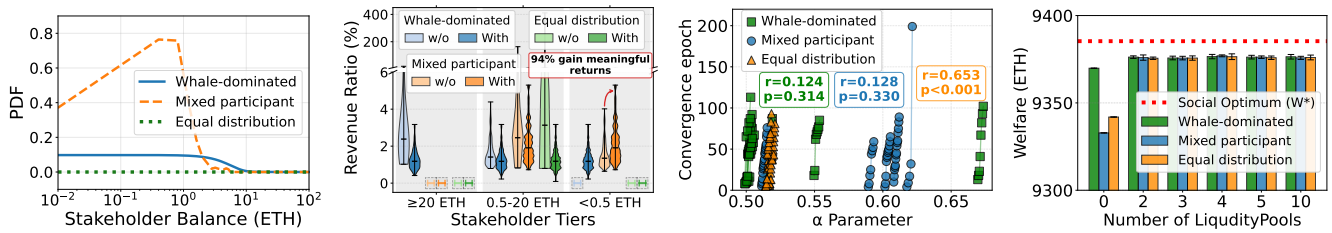
Input : Stakeholder v, Current epoch t, Perturbation parameter ϵ
Output : Participation choice $\mathcal{Y}_t(v)$ for epoch t

1 $\mathcal{D}_t \leftarrow \{D(b) \mid \forall b \in \mathcal{B}\}$
2 $\mathcal{V} \leftarrow$ All stakeholders join B2E
3 $\mathcal{F}_t \leftarrow$ B2E revenue in epoch t // *$\mathcal{F}_t(\overline{v})$ denotes the revenue of stakeholder $\overline{v} \in \mathcal{V}$ acting as an broker in B2E protocol.*
4 **if** $v \in \mathcal{V}$ **then**
5 $\quad F_e \leftarrow \mathcal{F}_t(v)$
6 **else**
7 $\quad \hat{\mathcal{V}} \leftarrow$ Sort $\mathcal{V}$ by $|D(v) - D(\overline{v})|, \forall \overline{v} \in \mathcal{V}$ in ascending order
8 $\quad$ Take first two stakeholders $v_1, v_2 \in \hat{\mathcal{V}}$
9 $\quad$ **if** v_1, v_2 *exist* **then**
10 $\quad\quad w_1 \leftarrow \frac{1}{|D_V(v)-D_V(v_1)|}$, $w_2 \leftarrow \frac{1}{|D_V(v)-D_V(v_2)|}$
11 $\quad\quad F_e \leftarrow \frac{w_1 \cdot \mathcal{F}_t(v_1) + w_2 \cdot \mathcal{F}_t(v_2)}{w_1 + w_2}$
12 $\quad$ **else**
13 $\quad\quad F_e \leftarrow D_V(v) \cdot \frac{\sum_{v \in \mathcal{V}} \mathcal{F}_t(v)}{\sum_{v \in \mathcal{V}} D_V(v)}$
14 $\mathcal{Y}_t(\overline{v}) = \{Y_b^{\overline{v}} = 0 \mid b \in \mathcal{B}\}, \overline{v} \in \mathcal{V}$
15 $F_b \leftarrow \max(\{F_t(b) + \epsilon \cdot \text{uniformRandom}(0, 1) \mid \forall b \in \mathcal{B}\})$
16 **if** $F_b > F_e$ **then**
17 $\quad \mathcal{Y}_t(v) \leftarrow \mathcal{Y}_t(v) \bigcup \{Y^v_{\arg\max(F_b)}\}$
18 **return** $\mathcal{Y}_t(v)$

5.2 Stakeholder Decision Mechanism

The goal of stakeholder participation is to maximize expected returns by choosing the optimal investment strategy among direct Broker2Earn participation and various LiquidityPool options. To achieve this goal, Algorithm 2 implements revenue comparisons and similarity-based estimation to guide stakeholder decision-making under incomplete information.

Algorithm 2 is executed epoch by epoch. Let t denote each epoch of the stakeholder decision system. At the beginning of epoch t, the algorithm observes market data including fund distributions $\mathcal{D}_t$, active stakeholder set $\mathcal{V}$, and revenue metrics $\mathcal{F}_t$ from Broker2Earn protocol interactions (lines 1-3). For stakeholders with historical participation data, direct revenue estimation uses previous performance $\mathcal{F}_t(v)$ (line 5). For new stakeholders lacking historical data, the algorithm implements similarity-based revenue estimation by sorting existing stakeholders by fund size proximity $|D_V(v) - D_V(\overline{v})|$ and selecting the two most similar investors (lines 7-8). Weighted interpolation using inverse distance weights produces revenue estimates for decision-making (lines 9-11). When there are insufficient similar stakeholders, the algorithm applies average revenue rates across all investors (line 13). The final participation decision compares the estimated direct participation revenue F_e with the maximum expected return from LiquidityPool options F_b, using a randomized perturbation (lines 14-16). The stakeholder selects the option with the higher expected return and implements the best-response strategies characterized in our theoretical analysis (line 17). The Algorithm 2 continues execution throughout the protocol's operational lifetime.



(a) PDF of different setups (b) Revenue across tiers (c) α effects on convergence (d) Welfare efficiency analysis

Figure 3: Experimental validation of theoretical predictions in ranking-dependent DeFi markets. Experiments demonstrate the inevitability of monopolization across market structures, configuration-dependent convergence patterns, and consistent welfare improvements enabled by LiquidityPool mechanisms.

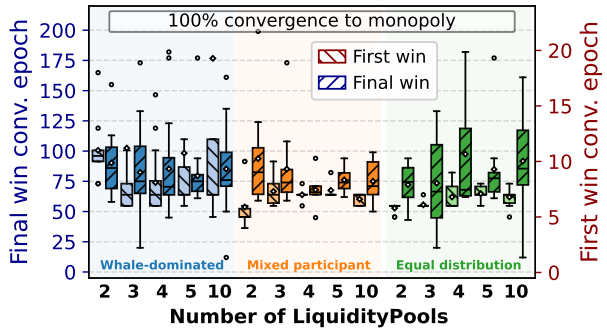


Figure 4: Convergence time distributions across market configurations. Box plots show First Win (initial monopolization) and Final Win (stable monopoly) for 2-10 competing LiquidityPools under three wealth distributions. All experiments converge to a monopoly (Theorem 5). Dual y-axes: Final Win 25-200 epochs (left), First Win 0-22 epochs (right).

6 Evaluation

6.1 Experimental Settings

Our experiments analyze real Ethereum blockchain transactions from blocks 19,000,000 to 19,999,999 using a Python-based analysis framework. The environment processes 200,000 cross-shard transactions per epoch over 300 epochs, with a 64-shard blockchain system (4 nodes per shard) supporting 1,000 stakeholders managing 1,000 ETH total.

We examine three stakeholder configurations designed to test knife-edge equilibrium predictions (Theorem 4) across different initial asymmetry levels (parameter α). The *whale-dominated* setup allocates 995 ETH to 60 large-volume stakeholders and 5 ETH to 940 small-volume stakeholders (individual balances less than 1 ETH), yielding a low α (high initial asymmetry). The *mixed participant* setup distributes 450 ETH to large users, 300 ETH to medium stakeholders, and 250 ETH to small-volume stakeholders, yielding a moderate α. The *equal distribution* setup allocates approximately 1 ETH per stakeholder, yielding $\alpha \approx \alpha^*$ (critical threshold). Each configuration is tested with 2, 3, 4, 5, and 10 competing LiquidityPools across 5 independent trials. We measure convergence time to monopolization, social welfare relative to the theoretical optimum W^*, and competitive dynamics between pools.

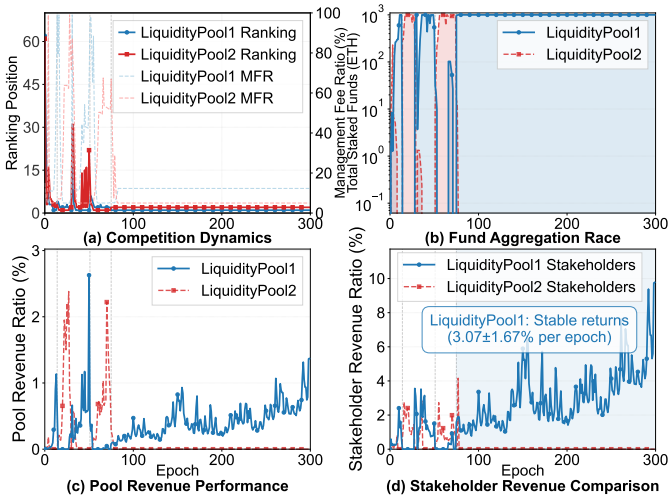


Figure 5: Competitive dynamics between two LiquidityPools demonstrating inevitable monopolization in ranking-dependent DeFi markets. The upper panels show LiquidityPool1 successfully achieving market dominance through strategic fee management and stakeholder attraction. Lower panels reveal LiquidityPool2's competitive failure, losing investors and revenue despite similar initial conditions, validating theoretical predictions of winner-takes-all market structures.

6.2 Monopolization Dynamics and Equilibrium Validation

Stakeholder configurations. Fig. 3(a) shows balance distributions across three configurations. The whale-dominated setup spans four orders of magnitude (0.001-400 ETH) with 60 large-volume stakeholders controlling 99.5% of funds. The mixed participant setup exhibits balances from 0.3 to 5.1 ETH across multiple tiers. The equal distribution setup allocates approximately 1.0 ETH per stakeholder.

Accessibility improvement. Fig. 3(b) demonstrates LiquidityPool's primary benefit: enabling participation for small-volume stakeholders. In the whale-dominated setup without LiquidityPool, minnows (<0.5 ETH) earn zero returns due to poor ranking positions. With LiquidityPool, 98% of minnows achieve median returns

of 1.18% per epoch through fund aggregation. This validates Theorem 1, which states that concentration dominates diversification in ranking-dependent protocols. Medium-tier stakeholders maintain stable returns across both conditions (1.41% vs 1.18%), while large-volume stakeholders experience reduced returns (2.38% vs 1.18%) due to increased competition from aggregated pools.

Universal monopolization. Fig. 4 shows convergence times across configurations and pool counts (2-10). All experiments converge to monopolistic equilibria with no exceptions, validating Theorem 5. We distinguish First Win (initial monopolization) and Final Win (stable monopoly). Median convergence times range from 25-50 epochs (First Win) and 50-75 epochs (Final Win) for equal distribution, to 75-100 and 100-125 epochs for whale-dominated setups. The gap between First and Final Win reflects algorithmic adjustments where pools temporarily lose dominance before re-establishing a monopoly.

Knife-edge equilibrium. Fig. 3(c) examines the relationship between initial asymmetry parameter α and convergence time across 15 experiments (3 setups $\times$ 5 trials, 2 pools each). Each trial generates multiple monopolization cycles, with initial α shown at the lowest point. Only equal distribution exhibits significant correlation ($r=0.653$, $p<0.001$), while whale-dominated and mixed setups show weak correlations. This confirms Theorem 4 that symmetric equilibria exist only when $\alpha \geq \alpha^*$.

Welfare comparison. Fig. 3(d) compares social welfare across pool counts. The social optimum, W^*, reaches 9,380 ETH. Direct participation $\tilde{W}$ achieves 9,330-9,340 ETH. Monopolistic equilibrium $\hat{W}$ achieves 9,360-9,375 ETH, validating the ranking $W^* > \hat{W} > \tilde{W}$(Theorem 6). The Price of Anarchy is 1.008, with welfare improvements of approximately 30 ETH over fragmented competition. This demonstrates that monopolistic concentration outperforms decentralized competition in ranking-dependent protocols.

6.3 Competitive Dynamics and Winner-Takes-All Validation

We examine a representative case from the whale-dominated setup with 2 LiquidityPools over 300 epochs Fig. 5) to illustrate monopolization mechanisms.

Strategic competition and ranking dynamics. Fig. 5(a) shows ranking positions (among 2 competing entities) and management fee ratios using a dual Y-axis. Pool1 experiences three distinct phases. Early phase (epochs 0-50): Pool1 struggles with volatile rankings (average rank 10, worst rank 62) and defensive high fees (up to 99% MFR). Middle phase (epochs 51-74): Pool1 establishes dominance (average rank 2.5) after competitor Pool2 drops to rank 13 at epoch 51. Final phase (epochs 75-300): Pool1 achieves a stable rank 1 monopoly with reduced MFR (12.3%). Pool2 follows complementary patterns, demonstrating zero-sum competition where one pool's ranking improvements directly cause declines in the other.

Fund concentration process. Fig. 5(b) tracks total staked funds. Pool1 captures 1000 ETH at epoch 11 but loses dominance at epoch 14 when participation drops from 100% to 0.2%. Pool2 similarly captures the entire market at epoch 17 but collapses at epoch 51 (participation falls from 94% to 1.5%). After epoch 75, Pool1 maintains a stable 100% participation while Pool2 drops to 0.1% (effectively zero). The fund dynamics create feedback: higher funds improve ranking through scale effects ($\partial^2 f/\partial D^2 > 0$), attracting more stakeholders and further increasing concentration, validating Lemma 1.

Pool operator outcomes. Fig. 5(c) presents pool revenue (operator profit from management fees). Pool1 achieves a stable revenue of 0.43% per epoch during the monopoly phase (epochs 75-300), despite a low MFR (12.3%), thanks to complete fund aggregation and superior ranking. Pool2 generates zero revenue after elimination, demonstrating that ranking-dependent protocols create winner-takes-all business models.

Stakeholder impact and accessibility. Fig. 5(d) shows stakeholder returns (after pool fees). During stable monopoly (epochs 75-300), Pool1 stakeholders achieve mean returns of 3.07% per epoch (median 2.66%, range 0.77-9.77%). Pool2 stakeholders receive zero returns after epoch 75. This outcome validates accessibility benefits established in Fig. 3(b): small-volume stakeholders who would earn near-zero returns through direct participation achieve competitive returns through fund aggregation. The time-series evidence confirms that 98% of minnows gain meaningful participation through LiquidityPool mechanisms, supporting Theorem 1, which posits that concentration dominates diversification in ranking-dependent protocols.

Zero-sum dynamics. The three turning points quantify competitive dynamics. Epoch 14: Pool2 dominates (rank 3 vs Pool1 rank 4, capturing 99.7% participation). Epoch 51: Pool1 overtakes (rank 3 vs Pool2 rank 13) through strategic MFR reduction. Epoch 75: Pool1 establishes a stable monopoly (rank 1, 93% participation) while Pool2 exits (7% participation $\rightarrow$0.1%). Stakeholder migration exhibits a near-perfect zero-sum property: epoch 74-75 shows that Pool1 gains 92.91% in participation while Pool2 loses 92.91%. This validates Theorem 5, which shows that ranking amplification creates winner-takes-all equilibria.

7 Conclusion

LiquidityPool, a fund aggregation protocol for ranking-dependent DeFi, establishes a theoretical framework that differs from traditional competition theory. Our knife-edge equilibrium analysis shows that symmetric multi-party equilibria exist only under restrictive initial conditions; otherwise, they converge to monopolistic structures. Our welfare analysis shows that monopolistic concentration outperforms fragmented competition in ranking-dependent protocols, with the social optimum achieving the highest welfare, unlike in the Bertrand and Cournot models. Experiments with real Ethereum transactions confirm universal convergence to a monopoly across wealth distributions and pool counts, while materially improving accessibility for small-volume stakeholders. These findings challenge standard antitrust intuitions by showing that fragmentation reduces welfare in ranking-dependent settings. Policy interventions should focus on fee regulation and governance mechanisms that balance scale benefits with contestability. Future work extends the framework to heterogeneous multi-protocol environments.

8 Acknowledgment

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A Proof

A.1 Proof of Theorem 1: All-or-Nothing Investment Strategy

Proof. We prove that concentrated allocation strategies weakly dominate diversified strategies under ranking-dependent revenue functions.

Consider stakeholder $v \in \mathcal{V}$ with fund endowment $D_V(v) > 0$ facing $|\mathcal{B}| + 1$ mutually exclusive investment options. Option 0 represents direct participation in Broker2Earn, yielding utility $u_0^v = f(D_V(v), r_v)$ where r_v denotes the stakeholder's ranking among direct investors. For each LiquidityPool $b \in \mathcal{B}$, option b yields utility $u_b^v = (1-\delta_b) \cdot \frac{D_V(v)}{D_B(b,\boldsymbol{\delta})} \cdot f(D_B(b,\boldsymbol{\delta}), r_b)$, where δ_b is the management fee, $D_B(b,\boldsymbol{\delta})$ represents total funds aggregated in pool b, and r_b denotes the pool's ranking position.

Any diversified allocation strategy $\boldsymbol{\varsigma}^v = (\varsigma_0^v, \varsigma_1^v, \ldots, \varsigma_{|\mathcal{B}|}^v)$ satisfying $\varsigma_0^v + \sum_{b\in\mathcal{B}} \varsigma_b^v = 1$ and $\varsigma_k^v \geq 0$ yields total utility:

$$U_f^v(\boldsymbol{\varsigma}^v) = \varsigma_0^v \cdot u_0^v + \sum_{b\in\mathcal{B}} \varsigma_b^v \cdot u_b^v = \sum_{k=0}^{|\mathcal{B}|} \varsigma_k^v \cdot u_k^v. \tag{8}$$

Define the concentrated strategy $\hat{\boldsymbol{\varsigma}}^v$ that allocates all funds to the option with maximum utility $k^* = \arg\max_{k\in\{0,1,\ldots,|\mathcal{B}|\}} u_k^v$. This concentrated strategy satisfies $\hat{\varsigma}_{k^*}^v = 1$ and $\hat{\varsigma}_k^v = 0$ for all $k \neq k^*$, yielding:

$$U_f^v(\hat{\boldsymbol{\varsigma}}^v) = u_{k^*}^v = \max_k u_k^v. \tag{9}$$

To establish dominance, observe that $u_{k^*}^v \geq u_k^v$ for all k. Using the normalization constraint:

$$U_f^v(\hat{\boldsymbol{\varsigma}}^v) = u_{k^*}^v = u_{k^*}^v \cdot 1 = u_{k^*}^v \cdot \left(\sum_{k=0}^{|\mathcal{B}|} \varsigma_k^v\right) \tag{10}$$

$$= \sum_{k=0}^{|\mathcal{B}|} \varsigma_k^v \cdot u_{k^*}^v \geq \sum_{k=0}^{|\mathcal{B}|} \varsigma_k^v \cdot u_k^v = U_f^v(\boldsymbol{\varsigma}^v), \tag{11}$$

where the inequality follows from $u_{k^*}^v \geq u_k^v$ and $\varsigma_k^v \geq 0$. Therefore, the concentrated strategy weakly dominates any diversified strategy.

The optimal option k^* can be characterized using the utility difference function. Comparing LiquidityPool b versus direct participation:

$$u_b^v - u_0^v = (1-\delta_b) \cdot \frac{D_V(v)}{D_B(b,\boldsymbol{\delta})} \cdot f(D_B(b,\boldsymbol{\delta}), r_b) - f(D_V(v), r_v) = h(v, \delta_b). \tag{12}$$

Stakeholder v should join pool b if $h(v,\delta_b) > 0$ and b offers the highest utility among all options. This completes the characterization of optimal concentrated strategies. □

A.2 Proof of Theorem 2: Stakeholder Best Response

Proof. We characterize the optimal participation strategy for a rational stakeholder facing competing LiquidityPool options and direct Broker2Earn participation.

Consider stakeholder $v \in \mathcal{V}$ with fund endowment $D_V(v) > 0$ who must choose among $|\mathcal{B}| + 1$ mutually exclusive options. The stakeholder's decision problem is:

$$\max_{\{Y_b^v\}} \sum_{b\in\mathcal{B}} Y_b^v \cdot (1-\delta_b) \cdot R(b,v) + \left(1 - \sum_{b\in\mathcal{B}} Y_b^v\right) \cdot f(D_V(v), r_v) \tag{13}$$

subject to $\sum_{b\in\mathcal{B}} Y_b^v \leq 1$ and $Y_b^v \in \{0,1\}$, where $R(b,v) = \frac{D_V(v)}{D_B(b,\boldsymbol{\delta})} \cdot f(D_B(b,\boldsymbol{\delta}), r_b)$ represents gross revenue from pool b.

The objective function can be rewritten by separating the direct participation baseline from incremental gains. For any pool b, the

incremental net revenue gain is:

$$\Delta U_b^v = Y_b^v \cdot (1-\delta_b) \cdot R(b,v) - Y_b^v \cdot f(D_V(v), r_v) \tag{14}$$

$$= Y_b^v \cdot \left[(1-\delta_b) \cdot \frac{D_V(v)}{D_B(b,\boldsymbol{\delta})} \cdot f(D_B(b,\boldsymbol{\delta}), r_b) - f(D_V(v), r_v) \right] \tag{15}$$

$$= Y_b^v \cdot h(v, \delta_b), \tag{16}$$

where the utility difference function is:

$$h(v,\delta_b) = (1-\delta_b) \cdot \frac{D_V(v)}{D_B(b,\boldsymbol{\delta})} \cdot f(D_B(b,\boldsymbol{\delta}), r_b) - f(D_V(v), r_v). \tag{17}$$

The stakeholder's optimization reduces to choosing which pool maximizes $\Delta U_b^v = Y_b^v \cdot h(v,\delta_b)$ subject to $\sum_b Y_b^v \le 1$ and $Y_b^v \in \{0,1\}$. Identify the pool offering the highest utility difference:

$$b' = \arg\max_{b \in \mathcal{B}} h(v, \delta_b). \tag{18}$$

We now characterize the optimal strategy through case analysis:

Case 1: $h(v,\delta_{b'}) > 0$**.** The best pool offers strictly positive incremental utility. Joining pool b' yields total utility $f(D_V(v), r_v) + h(v,\delta_{b'}) > f(D_V(v), r_v)$, strictly exceeding direct participation. The optimal strategy is $Y_{b'}^{v*} = 1$ and $Y_b^{v*} = 0$ for all $b \neq b'$.

Case 2: $h(v,\delta_{b'}) < 0$**.** Even the best pool offers negative incremental utility. All pools yield lower net revenue than direct participation. The optimal strategy is $Y_b^{v*} = 0$ for all $b \in \mathcal{B}$, corresponding to direct participation. The stakeholder's utility is $f(D_V(v), r_v)$, which exceeds $f(D_V(v), r_v) + h(v,\delta_b)$ for all b.

Case 3: $h(v,\delta_{b'}) = 0$**.** The best pool offers exactly zero incremental utility. Both joining pool b' and participating directly yield identical net revenue $f(D_V(v), r_v)$. The optimal strategy can be either $Y_{b'}^{v*} = 1$ (with $Y_b^{v*} = 0$ for $b \neq b'$) or $Y_b^{v*} = 0$ for all b, denoted by $Y_{b'}^{v*} \in \{0,1\}$.

Combining these cases completes the characterization stated in the theorem. □

A.3 Proof of Lemma 1: Fund Migration Properties

Proof. We establish the two properties separately.

Proof of Property (1): Non-increasing Step Function.

The step function structure arises from discrete stakeholder decisions. Each stakeholder $v \in \mathcal{V}$ makes binary decisions $Y_b^v \in \{0,1\}$ based on Theorem 2. While δ_b and $h(v,\delta_b)$ vary continuously, decisions Y_b^v change only at finitely many threshold points where $h(v,\delta_b)$ crosses zero, or argmax conditions change.

Between consecutive threshold points, all stakeholder decisions remain fixed. Therefore:

$$D_B(b,\boldsymbol{\delta}) = \sum_{v \in \mathcal{V}} Y_b^v \cdot D_V(v) \tag{19}$$

remains constant as δ_b varies. Since there are finitely many stakeholders $|\mathcal{V}| < \infty$ and finitely many pools $|\mathcal{B}| < \infty$, finitely many threshold points exist, establishing that $D_B(b,\boldsymbol{\delta})$ is a step function.

For monotonicity, we prove by contradiction. Suppose there exist fee ratios $\delta_b' < \delta_b''$ such that:

$$D_B(b,\boldsymbol{\delta}|_{\delta_b=\delta_b'}) < D_B(b,\boldsymbol{\delta}|_{\delta_b=\delta_b''}). \tag{20}$$

This requires at least one stakeholder $\hat{v}$ to switch from not joining pool b at lower fee δ_b' to joining at higher fee δ_b''. By Theorem 2, $\hat{v}$ joins pool b only when $h(\hat{v},\delta_b) > 0$ and b offers highest utility.

However, the utility difference is:

$$h(\hat{v},\delta_b) = (1-\delta_b) \cdot \frac{D_V(\hat{v})}{D_B(b,\boldsymbol{\delta})} \cdot f(D_B(b,\boldsymbol{\delta}), r_b) - f(D_V(\hat{v}), r_{\hat{v}}). \tag{21}$$

Since $\delta_b' < \delta_b''$, we have $(1-\delta_b') > (1-\delta_b'')$. The utility at the lower fee satisfies:

$$h(\hat{v},\delta_b') = (1-\delta_b') \cdot \frac{D_V(\hat{v})}{D_B(b,\boldsymbol{\delta}|_{\delta_b=\delta_b'})} \cdot f(D_B(b,\boldsymbol{\delta}|_{\delta_b=\delta_b'}), r_b') - f(D_V(\hat{v}), r_{\hat{v}}) \tag{22}$$

$$> (1-\delta_b'') \cdot \frac{D_V(\hat{v})}{D_B(b,\boldsymbol{\delta}|_{\delta_b=\delta_b'})} \cdot f(D_B(b,\boldsymbol{\delta}|_{\delta_b=\delta_b'}), r_b') - f(D_V(\hat{v}), r_{\hat{v}}) \tag{23}$$

$$\ge (1-\delta_b'') \cdot \frac{D_V(\hat{v})}{D_B(b,\boldsymbol{\delta}|_{\delta_b=\delta_b''})} \cdot f(D_B(b,\boldsymbol{\delta}|_{\delta_b=\delta_b''}), r_b'') - f(D_V(\hat{v}), r_{\hat{v}}) \tag{24}$$

$$= h(\hat{v},\delta_b''), \tag{25}$$

where the second inequality holds because increasing δ_b can only decrease or maintain $D_B(b,\boldsymbol{\delta})$ (by the property we are proving), potentially worsening ranking r_b. Both effects weakly decrease the utility difference.

Therefore, if $h(\hat{v},\delta_b') \le 0$ or pool b was not the argmax at a lower fee, the same holds at a higher fee, contradicting that $\hat{v}$ switches to joining b at a higher fee. We conclude $D_B(b,\boldsymbol{\delta})$ is non-increasing.

Proof of Property (2): Strategic Substitutability.

LiquidityPool b's revenue is $U_l^b(\boldsymbol{\delta}) = \delta_b \cdot f(D_B(b,\boldsymbol{\delta}), r_b)$. Consider competitor $b' \neq b$ increasing fee from $\delta_{b'}$ to $\delta_{b'} + \epsilon$ for small $\epsilon > 0$. By Theorem 2, stakeholders evaluate participation based on utility differences. When b' raises fees:

$$h(v,\delta_{b'}+\epsilon) = (1-\delta_{b'}-\epsilon) \cdot \frac{D_V(v)}{D_B(b',\boldsymbol{\delta}')} \cdot f(D_B(b',\boldsymbol{\delta}'), r_{b'}) - f(D_V(v), r_v), \tag{26}$$

which is strictly less than $h(v,\delta_{b'})$ due to reduced retention factor $(1-\delta_{b'}-\epsilon) < (1-\delta_{b'})$.

Some stakeholders who marginally preferred b' now find $h(v,\delta_{b'}+\epsilon) < 0$ or alternative pools more attractive. Let $\mathcal{S}_{b \leftarrow b'}(\epsilon)$ denote stakeholders migrating from b' to b:

$$\Delta D_B(b) = \sum_{v \in \mathcal{S}_{b \leftarrow b'}(\epsilon)} D_V(v) > 0. \tag{27}$$

This migration creates two reinforcing effects. First, scale effects from a larger fund base increase revenue through convexity $\frac{\partial^2 f}{\partial D^2} > 0$:

$$f(D_B(b,\boldsymbol{\delta}) + \Delta D_B(b), r_b) > f(D_B(b,\boldsymbol{\delta}), r_b) + \frac{\partial f}{\partial D}\Big|_{D=D_B(b,\boldsymbol{\delta})} \cdot \Delta D_B(b). \tag{28}$$

Second, ranking effects from a higher market share may improve pool b's ranking from r_b to $r_b' < r_b$ (better position). By Eq. (3), better rankings yield higher revenue: $f(D, r_b') > f(D, r_b)$ when $r_b' < r_b$. Both effects increase pool b's total revenue base $f(D_B(b,\boldsymbol{\delta}), r_b)$.

The strategic substitutability arises because these effects make higher fees δ_b more profitable. Before competitor b' raised fees, pool b's marginal revenue from increasing δ_b balanced direct fee

gain against fund loss from stakeholder departures. After b''s fee increase, pool b gains market share and improves its ranking, expanding its revenue base. Moreover, stakeholders migrating from b' are less price-sensitive to b's moderate fee increases, as long as b remains more attractive than b'. This reduces the fund loss rate $|\frac{\partial D_B(b,\boldsymbol{\delta})}{\partial \delta_b}|$, making fee increases more profitable.

Formally, the cross-partial derivative captures this:

$$\frac{\partial^2 U_l^b}{\partial \delta_b \partial \delta_{b'}} = \frac{\partial}{\partial \delta_{b'}} \left(\frac{\partial U_l^b}{\partial \delta_b} \right) < 0, \tag{29}$$

where the negative sign indicates that increases in $\delta_{b'}$ increase the marginal profitability of increasing δ_b, establishing strategic substitutability. □

A.4 Proof of Theorem 3: Nash Equilibrium Existence

Proof. We establish existence using Brouwer's fixed point theorem despite step-function fund dynamics.

The strategy space $\Delta = \prod_{b \in \mathcal{B}} [0,1]$ is compact and convex. Stakeholders make binary decisions $\mathcal{Y}^v = \{Y_b^v \mid b \in \mathcal{B}\}$ with $\sum_b Y_b^v \leq 1$ and $Y_b^v \in \{0,1\}$. By Theorem 2, given any $\boldsymbol{\delta}$, stakeholders have well-defined best responses determined by utility differences $h(v, \delta_b)$.

For each pool $b \in \mathcal{B}$, define best response correspondence:

$$R_b(\boldsymbol{\delta}_{-b}) = \arg\max_{\delta_b \in [0,1]} U_l^b(\{\delta_b\} \cup \boldsymbol{\delta}_{-b}, \mathcal{Y}(\{\delta_b\} \cup \boldsymbol{\delta}_{-b})), \tag{30}$$

where $\mathcal{Y}(\boldsymbol{\delta})$ denotes induced stakeholder participation.

By Lemma 1, $D_B(b, \boldsymbol{\delta})$ is a step function taking finitely many constant values. Revenue $U_l^b = \delta_b \cdot f(D_B(b, \boldsymbol{\delta}), r_b)$ exhibits piecewise continuity: continuous on intervals where D_B is constant, with downward jumps at thresholds preserving upper semi-continuity. The extreme value theorem guarantees $R_b(\boldsymbol{\delta}_{-b})$ is non-empty for all $\boldsymbol{\delta}_{-b}$.

For upper semi-continuity of R_b in $\boldsymbol{\delta}_{-b}$, consider sequence $\boldsymbol{\delta}_{-b}^{(n)} \to \boldsymbol{\delta}_{-b}$ with optimal responses $\delta_b^{(n)} \in R_b(\boldsymbol{\delta}_{-b}^{(n)})$. Discontinuities in stakeholder participation occur only when utility differences $h(v, \delta_b)$ cross zero or argmax conditions change—events forming a measure-zero set in $\boldsymbol{\delta}_{-b}$ space. Between discontinuity points, U_l^b varies continuously. Any convergent subsequence $\delta_b^{(n_k)} \to \bar{\delta}_b$ satisfies:

$$U_l^b(\{\bar{\delta}_b\} \cup \boldsymbol{\delta}_{-b}, \mathcal{Y}(\{\bar{\delta}_b\} \cup \boldsymbol{\delta}_{-b})) \geq \limsup_{k \to \infty} U_l^b(\{\delta_b^{(n_k)}\} \cup \boldsymbol{\delta}_{-b}^{(n_k)}, \mathcal{Y}^{(n_k)}) \tag{31}$$

by upper semi-continuity. Since $\delta_b^{(n_k)}$ was optimal:

$$U_l^b(\{\delta_b^{(n_k)}\} \cup \boldsymbol{\delta}_{-b}^{(n_k)}, \mathcal{Y}^{(n_k)}) \geq U_l^b(\{\delta_b\} \cup \boldsymbol{\delta}_{-b}^{(n_k)}, \mathcal{Y}(\{\delta_b\} \cup \boldsymbol{\delta}_{-b}^{(n_k)})) \tag{32}$$

for all $\delta_b \in [0,1]$. Taking limits establishes $\bar{\delta}_b \in R_b(\boldsymbol{\delta}_{-b})$, confirming upper semi-continuity.

Define joint correspondence $R(\boldsymbol{\delta}) = (R_1(\boldsymbol{\delta}_{-1}), \ldots, R_{|\mathcal{B}|}(\boldsymbol{\delta}_{-|\mathcal{B}|})) : \Delta \rightrightarrows \Delta$. Since R is non-empty valued and upper semi-continuous on compact convex Δ, Brouwer's theorem guarantees fixed point $\boldsymbol{\delta}^* \in R(\boldsymbol{\delta}^*)$, meaning $\delta_b^* \in R_b(\boldsymbol{\delta}_{-b}^*)$ for all b.

This fixed point satisfies:

$$U_l^b(\{\delta_b^*\} \cup \boldsymbol{\delta}_{-b}^*, \mathcal{Y}^*) \geq U_l^b(\{\delta_b\} \cup \boldsymbol{\delta}_{-b}^*, \mathcal{Y}^*), \quad \forall \delta_b \in [0,1],\ b \in \mathcal{B}, \tag{33}$$

where $\mathcal{Y}^* = \mathcal{Y}(\boldsymbol{\delta}^*)$. By Theorem 2, $\mathcal{Y}^*$ represents stakeholder best responses, establishing $(\boldsymbol{\delta}^*, \mathcal{Y}^*)$ as Nash equilibrium. □

A.5 Proof of Theorem 4: Knife-Edge Equilibrium Property

Proof. We establish knife-edge property through linearized stability analysis. In symmetric equilibrium, all pools set $\delta_b^* = \delta^*$ and receive $D_B(b, \boldsymbol{\delta}^*) = \frac{D_{\text{total}}}{|\mathcal{B}|}$ where $D_{\text{total}} = \sum_v D_V(v)$.

Define deviation vector $\mathbf{x}(t) = (x_1(t), \ldots, x_{|\mathcal{B}|}(t))$ where $x_b(t) = D_B(b, \boldsymbol{\delta}^t) - \frac{D_{\text{total}}}{|\mathcal{B}|}$ measures fund deviation. Normalization implies $\sum_b x_b(t) = 0$. Asymmetry parameter α bounds initial deviations: $\max_b |x_b(0)| \leq \frac{D_{\text{total}}}{|\mathcal{B}|} \cdot \frac{1-\alpha}{\alpha}$.

Linearized dynamics are $\frac{dx_b(t)}{dt} = \sum_{b'} A_{bb'} x_{b'}(t)$ where stability matrix elements are:

$$A_{bb'} = \begin{cases} -\lambda(\frac{\partial^2 f}{\partial D^2} + \frac{\partial^2 f}{\partial D \partial r}), & \text{if } b = b', \\ \mu \frac{\partial^2 f}{\partial D^2}, & \text{if } b \neq b', \end{cases} \tag{34}$$

with $\lambda, \mu > 0$ as migration response parameters. Diagonal terms represent self-stabilization (outflow when relatively unattractive) while off-diagonal terms represent competitive amplification (inflow from competitors).

For symmetric structure with constant diagonal a and off-diagonal c, eigenvalues are $\lambda_1 = a + (|\mathcal{B}| - 1)c$ and $\lambda_2 = \cdots = \lambda_{|\mathcal{B}|} = a - c$. Since deviations satisfy $\sum_b x_b = 0$, relevant eigenvalue is:

$$\lambda_{\max} = a - c = -\lambda \frac{\partial^2 f}{\partial D^2} + (|\mathcal{B}| - 1)\mu \frac{\partial^2 f}{\partial D^2} - \lambda \frac{\partial^2 f}{\partial D \partial r}. \tag{35}$$

Stability requires $\lambda_{\max} < 0$, yielding:

$$(|\mathcal{B}| - 1)\mu - \lambda < \lambda \cdot \frac{\partial^2 f / \partial D \partial r}{\partial^2 f / \partial D^2}. \tag{36}$$

Critical threshold α^* where $\lambda_{\max} = 0$ satisfies:

$$\alpha^* = \left[1 + \frac{(|\mathcal{B}| - 1)\mu - \lambda}{\lambda} \cdot \frac{\partial^2 f / \partial D^2}{\partial^2 f / \partial D \partial r} \right]^{-1}. \tag{37}$$

When $\alpha \geq \alpha^*$, initial perturbations ensure $\lambda_{\max} < 0$ (exponential decay). When $\alpha < \alpha^*$, positive $\lambda_{\max} > 0$ causes exponential divergence: $\|\mathbf{x}(t)\| \sim e^{\lambda_{\max} t} \|\mathbf{x}(0)\|$, amplifying asymmetries through strategic substitutability. □

A.6 Proof of Theorem 5: Market Structure and Monopolization

Proof. We establish monopolization through Lyapunov stability analysis. By Theorem 4, symmetric equilibria do not exist when $\alpha < \alpha^*$. Define Lyapunov function $V(t) = \sum_{b \in \mathcal{B}} [D_B(b, \boldsymbol{\delta}^t)]^2$ measuring fund concentration, bounded above by $(\sum_v D_V(v))^2$.

Time derivative along trajectories is:

$$\frac{dV(t)}{dt} = 2 \sum_{b \in \mathcal{B}} D_B(b, \boldsymbol{\delta}^t) \cdot g_b(\mathbf{D}^t, \mathbf{r}^t), \tag{38}$$

where $g_b(\mathbf{D}, \mathbf{r})$ denotes net migration rate to pool b.

Consider pools b_1, b_2 with $D_B(b_1, \boldsymbol{\delta}) > D_B(b_2, \boldsymbol{\delta}) > 0$. By ranking rules, $r_{b_1} < r_{b_2}$ (better position). Revenue per unit satisfies:

$$\frac{f(D_B(b_1, \boldsymbol{\delta}), r_{b_1})}{D_B(b_1, \boldsymbol{\delta})} > \frac{f(D_B(b_2, \boldsymbol{\delta}), r_{b_2})}{D_B(b_2, \boldsymbol{\delta})} \tag{39}$$

due to convexity ($\frac{\partial^2 f}{\partial D^2} > 0$) and ranking effects. This drives migration $g_{b_1} > 0$ and $g_{b_2} < 0$. The key inequality is:

$$D_B(b_1, \boldsymbol{\delta}) \cdot g_{b_1} + D_B(b_2, \boldsymbol{\delta}) \cdot g_{b_2} = (D_B(b_1, \boldsymbol{\delta}) - D_B(b_2, \boldsymbol{\delta})) \cdot \Delta > 0 \tag{40}$$

where Δ denotes migration amount. Summing over all pairs establishes $\frac{dV(t)}{dt} > 0$ whenever multiple pools coexist.

Since $V(t)$ is bounded and strictly increasing, it converges. Derivative $\frac{dV(t)}{dt} = 0$ occurs only when $g_b = 0$ for all active pools, happening only at corners where one pool holds all funds. Monopolist identity is $b^* = \arg\max_b D_B(b, \boldsymbol{\delta}^0)$, yielding:

$$\lim_{t\to\infty} D_B(b^*, \boldsymbol{\delta}^t) = \sum_v D_V(v), \quad \lim_{t\to\infty} D_B(b, \boldsymbol{\delta}^t) = 0, \quad \forall b \neq b^*. \tag{41}$$

For no intermediate equilibria, suppose asymmetric configuration with $D_B(b_1, \boldsymbol{\delta}^*) > D_B(b_2, \boldsymbol{\delta}^*) > 0$. If $\delta^*_{b_1} \leq \delta^*_{b_2}$, pool b_1 dominates on all dimensions, contradicting equilibrium as all stakeholders prefer b_1. If $\delta^*_{b_1} > \delta^*_{b_2}$, pool b_2 can profitably deviate by reducing fees to attract direct investors with worse rankings, contradicting the Nash equilibrium. Therefore, only symmetric or monopolistic equilibria exist. □

A.7 Proof of Theorem 6: Welfare Analysis

Proof. We establish the welfare ordering $W^* > \hat{W} > \tilde{W}$ by separately proving each inequality using convexity and ranking effects.

Social optimum welfare achieves complete fund aggregation at rank 1 with zero fees: $W^* = f(D_{\max}, 1)$ where $D_{\max} = \sum_{v\in\mathcal{V}} D_V(v)$.

Monopolistic equilibrium welfare when $\alpha < \alpha^*$ involves one dominant LiquidityPool b^* charging fee $\hat{\delta} > 0$ at ranking $\hat{r}$, plus excluded stakeholders $\mathcal{M}$: $\hat{W} = f(D_B(b^*, \hat{\boldsymbol{\delta}}), \hat{r}) + \sum_{v\in\mathcal{M}} f(D_V(v), r_v)$ where $D_B(b^*, \hat{\boldsymbol{\delta}}) = \sum_{v\in\mathcal{V}\setminus\mathcal{M}} D_V(v)$.

Symmetric equilibrium welfare when $\alpha \geq \alpha^*$ fragments funds across $|\mathcal{B}|$ pools: $\tilde{W} = \sum_{b\in\mathcal{B}} f(D_B(b, \tilde{\boldsymbol{\delta}}), \tilde{r}_b) + \sum_{v\in\mathcal{M}} f(D_V(v), r_v)$ where $D_B(b, \tilde{\boldsymbol{\delta}}) = \frac{D_{\text{pool}}}{|\mathcal{B}|}$ with $D_{\text{pool}} = \sum_{v\in\mathcal{V}\setminus\mathcal{M}} D_V(v)$.

Proof of $W^* > \hat{W}$: Let $D_{\text{excl}} = \sum_{v\in\mathcal{M}} D_V(v)$ denote excluded funds, so $D_B(b^*, \hat{\boldsymbol{\delta}}) = D_{\max} - D_{\text{excl}}$. Excluded stakeholders receive inferior rankings $r_v > \hat{r}$ compared to the pool's ranking. By convexity $\frac{\partial^2 f}{\partial D^2} > 0$ and ranking dependence, aggregating excluded funds with the pool yields:

$$\begin{aligned} f(D_{\max}, 1) &> f(D_{\max} - D_{\text{excl}}, \hat{r}) + f(D_{\text{excl}}, \min_{v\in\mathcal{M}} r_v) \\ &\geq f(D_{\max} - D_{\text{excl}}, \hat{r}) + \sum_{v\in\mathcal{M}} f(D_V(v), r_v) = \hat{W}, \end{aligned} \tag{42}$$

where the first inequality uses convexity and improved ranking from complete aggregation, and the second uses convexity applied to fragmented excluded funds. Therefore, $W^* > \hat{W}$.

Proof of $\hat{W} > \tilde{W}$: Let D_{pool} denote total pooled funds. Monopolistic equilibrium concentrates all in b^*: $D_B(b^*, \hat{\boldsymbol{\delta}}) = D_{\text{pool}}$ achieving ranking $\hat{r}$. Symmetric equilibrium fragments equally: $D_B(b, \tilde{\boldsymbol{\delta}}) = \frac{D_{\text{pool}}}{|\mathcal{B}|}$ for each $b \in \mathcal{B}$, achieving rankings $\tilde{r}_b \approx \frac{|\mathcal{B}|+1}{2}$.

By Jensen's inequality for convex function $f(\cdot, r)$, we have $f(D_{\text{pool}}, r) > |\mathcal{B}| \cdot f\left(\frac{D_{\text{pool}}}{|\mathcal{B}|}, r\right)$ for any fixed ranking r. Additionally, monopolistic concentration achieves a superior ranking $\hat{r} < \tilde{r}_b$ due to a larger fund size. Combining both effects:

$$f(D_{\text{pool}}, \hat{r}) > f(D_{\text{pool}}, \tilde{r}_b) > |\mathcal{B}| \cdot f\left(\frac{D_{\text{pool}}}{|\mathcal{B}|}, \tilde{r}_b\right) \geq \sum_{b\in\mathcal{B}} f\left(\frac{D_{\text{pool}}}{|\mathcal{B}|}, \tilde{r}_b\right), \tag{43}$$

where the first inequality uses ranking effects, the second uses convexity, and the final inequality holds with equality when all symmetric pools have identical rankings.

Since direct investor contributions $\sum_{v\in\mathcal{M}} f(D_V(v), r_v)$ are identical in both scenarios, we have:

$$\begin{aligned} \hat{W} &= f(D_{\text{pool}}, \hat{r}) + \sum_{v\in\mathcal{M}} f(D_V(v), r_v) \\ &> \sum_{b\in\mathcal{B}} f\left(\frac{D_{\text{pool}}}{|\mathcal{B}|}, \tilde{r}_b\right) + \sum_{v\in\mathcal{M}} f(D_V(v), r_v) = \tilde{W}. \end{aligned} \tag{44}$$

Combining both inequalities establishes $W^* > \hat{W} > \tilde{W}$. The Price of Anarchy when $\alpha < \alpha^*$ is $\text{PoA}(\alpha) = \frac{W^*}{\hat{W}}$, bounded by $\gamma > 1$ representing maximum exclusion. When $\alpha \geq \alpha^*$, the PoA is $\frac{W^*}{\tilde{W}}$, measuring efficiency loss from fund fragmentation. □